1 Solution — GIAM 1.1 Problem 9

1.1 Problem

Perform the following computations with complex numbers:

- **(a)** $(4+3i)-(3+2i)$
- **(b)** $(1+i)+(1-i)$
- **(c)** $(1+i)\cdot(1-i)$
- **(d)** $(2-3i)\cdot(3-2i)$

1.2 Setup — Rules of Complex Arithmetic

For $z=a+bi$ and $w=c+di$ with $a,b,c,d\in\mathbb{R}$:

- **Addition** / **subtraction** is component-wise:

$$z \pm w \;=\; (a\pm c)+(b\pm d)\,i.$$

- **Multiplication** uses FOIL, then collapses $i^2=-1$:

$$z\cdot w \;=\; (a+bi)(c+di) \;=\; ac+adi+bci+bd\,i^2 \;=\; (ac-bd)+(ad+bc)\,i.$$

That's all that's needed. Now the four computations.

1.3 Part (a) — $(4+3i)-(3+2i)$

Component-wise subtraction:

$$(4+3i)-(3+2i) \;=\; (4-3)+(3-2)\,i \;=\; \boxed{1+i}.$$

1.4 Part (b) — $(1+i)+(1-i)$

Component-wise addition. The imaginary parts $+i$ and $-i$ cancel:

$$(1+i)+(1-i) \;=\; (1+1)+(1-1)\,i \;=\; \boxed{2}.$$

The result is a *real* number — the imaginary part vanished.

1.5 Part (c) — $(1+i)\cdot(1-i)$

FOIL:

$$\begin{aligned}(1+i)(1-i) &= 1\cdot 1+1\cdot(-i)+i\cdot 1+i\cdot(-i)\\ &= 1 \;-\; i \;+\; i \;-\; i^2\\ &= 1 \;-\; (-1)\\ &= \boxed{2}.\end{aligned}$$

Faster route — recognize the **difference-of-squares** pattern:

$$(1+i)(1-i) \;=\; 1^2 \;-\; i^2 \;=\; 1 \;-\; (-1) \;=\; 2.$$

(The pair $z = 1+i$ and $\bar{z} = 1-i$ are **complex conjugates**; their product is $|z|^2 = 1^2+1^2 = 2$.)

1.6 Part (d) — $(2-3i)\cdot(3-2i)$

FOIL again:

$$\begin{aligned}(2-3i)(3-2i) &= 2\cdot 3+2\cdot(-2i)+(-3i)\cdot 3+(-3i)\cdot(-2i)\\ &= 6 \;-\; 4i \;-\; 9i \;+\; 6\,i^2\\ &= 6 \;-\; 13i \;+\; 6\cdot(-1)\\ &= (6-6)+(-13)\,i\\ &= \boxed{-13i}.\end{aligned}$$

The result is a *purely imaginary* number — the real parts cancelled.

1.7 Visualizing (a) and (b) Geometrically

Complex addition/subtraction is *vector* arithmetic in the plane: $z = a+bi$ corresponds to the point (or arrow from the origin to the point) $(a,b) \in \mathbb{R}^2$.

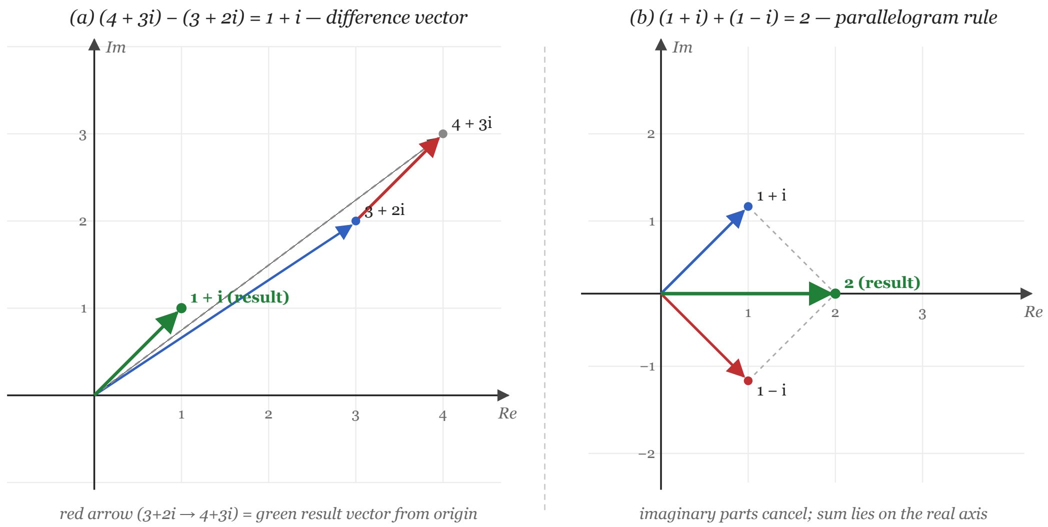

Figure 1.1: Left: (4+3i) − (3+2i) = 1+i is the vector from the tip of 3+2i to the tip of 4+3i, which equals the green result vector from the origin. Right: (1+i) + (1−i) = 2 is the diagonal of the parallelogram with sides 1+i and 1−i; the imaginary parts cancel so the sum lies on the real axis.

- **(a)**: subtraction $z - w$ is the *vector from the tip of* w *to the tip of* z. In the figure, $(3 + 2i) \to (4 + 3i)$ is one step right and one step up — exactly the vector $1 + i$. Translated to the origin, that's the result vector.
- **(b)**: addition $z + w$ obeys the *parallelogram rule*. Here $z = 1 + i$ and $w = 1 - i$ are mirror images across the real axis, so the parallelogram is symmetric and the sum lies on the real axis: imaginary parts annihilate, real parts add to 2.

1.8 Summary

Part	Computation	Result
(a)	$(4 + 3i) - (3 + 2i)$	$1 + i$
(b)	$(1 + i) + (1 - i)$	2
(c)	$(1 + i)(1 - i)$	2
(d)	$(2 - 3i)(3 - 2i)$	$-13i$

Table 1.1

1.9 Remark — Conjugates and the Modulus

Item (c) is a special case of a *very* useful identity:

$$z \cdot \bar{z} \;=\; |z|^2 \;\in\; \mathbb{R}_{\geq 0},$$

where $z = a + bi$ and $\bar{z} = a - bi$ is the **complex conjugate**. For $z = 1 + i$: $\bar{z} = 1 - i$, and $z\bar{z} = 1^2 + 1^2 = 2$.

This identity is the engine behind *complex division*: to compute z/w, multiply numerator and denominator by $\bar{w}$:

$$\frac{z}{w} \;=\; \frac{z\,\bar{w}}{w\,\bar{w}} \;=\; \frac{z\,\bar{w}}{|w|^2}.$$

The denominator becomes a real number; the rest is just FOIL. For instance: $1/(1 + i) = (1 - i)/2 = \frac{1}{2} - \frac{1}{2}i$.

1.10 Remark — Why (d) Lands Purely Imaginary

The product in (d) wasn't a coincidence — it was a *near*-conjugate situation in disguise:

$$(2 - 3i) \;=\; -i \cdot (3 + 2i)$$

(check: $-i \cdot (3 + 2i) = -3i - 2i^2 = -3i + 2 = 2 - 3i$. ✓)

So

$$(2 - 3i)(3 - 2i) \;=\; -i \cdot (3 + 2i)(3 - 2i) \;=\; -i \cdot |3 + 2i|^2 \;=\; -i \cdot (9 + 4) \;=\; -13i.$$

The factor $(3 + 2i)(3 - 2i) = 13$ — the squared modulus — and the leading $-i$ rotates by -90°, landing the answer on the negative imaginary axis.

1.11 Remark — Multiplication as Rotation Plus Scaling

For (c) and (d), there's a clean *geometric* interpretation. Writing complex numbers in **polar form** $z = re^{i\theta}$ (where $r = |z|$ and $\theta = \arg z$), multiplication satisfies

$$z_1 \cdot z_2 \;=\; (r_1 r_2)\, e^{i(\theta_1+\theta_2)}.$$

Moduli multiply, arguments add. Applying this to (c) and (d):

product	factors' moduli	factors' arguments	product modulus	product argument	rectangular
$(1+i)(1-i)$	$\sqrt{2}\cdot\sqrt{2}=2$	$+\pi/4+(-\pi/4)=0$	2	0	2
$(2-3i)(3-2i)$	$\sqrt{13}\cdot\sqrt{13}=13$	$\approx -56.3^\circ+(-33.7^\circ)=-90^\circ$	13	$-\pi/2$	$-13i$

Table 1.2

So (c)'s factors are at angles $\pm 45^\circ$, summing to 0° — the product lies on the positive real axis. And (d)'s factors are at angles roughly -56° and -34°, summing to exactly -90° — the product lies on the negative imaginary axis.

This explains the "miraculous" cancellation of imaginary parts in (c) and real parts in (d): the geometry is doing the work.

1.12 Remark — All Four Operations Stay Inside $\mathbb{C}$

The four computations illustrate that $\mathbb{C}$ is closed under addition, subtraction, and multiplication (and division, when the divisor is nonzero). In each case the result is again of the form $\text{real} + \text{imag}\cdot i$ with real components — never anything "more exotic" needed. This is the closure property that makes $\mathbb{C}$ a *field*, extending the field $\mathbb{R}$.

The price paid for this closure is: $\mathbb{C}$ has no consistent *order* — 1 vs. i has no natural "bigger or smaller" (this is the textbook's "the complex numbers can't really be put into a well-defined order" remark). We trade ordering for algebraic closure and the existence of $\sqrt{-1}$.